AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-0841	**Classification:**	Major / Critical Quality Attribute (CQA)
Product / API Name:	Atorvastatin Calcium Trihydrate API	**Batch / Lot Number:**	ATC-2026-B09
Manufacturing Stage:	Stage 4: Crystallization, Filtration & Vacuum Drying	**Equipment Tag:**	Reactor CRYS-RX-04 / ANFD-02
Date of Occurrence:	2026-10-22 14:15 UTC	**Date of Discovery:**	2026-10-23 09:30 UTC
Material Lot ID:	Ethanol Wash RAW-ETH-4029	**Reporting Department:**	Quality Control Analytical Laboratory

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Chromatographic Purity: Related Substance Impurity D	**NMT 0.15% (w/w)**	**0.28% (w/w) [OUT OF SPECIFICATION]**	USP-NF / HPLC (Waters Alliance #HPLC-8812)
Assay (Anhydrous Basis)	98.0% - 102.0%	99.4%	HPLC UV Detection at 246 nm
Water Content (Karl Fischer)	3.5% - 5.5%	4.1%	Mettler Toledo V30S Titrator
Residual Ethanol	NMT 5000 ppm	1820 ppm	Headspace GC-FID

2. INCIDENT NARRATIVE & CHRONOLOGY

During release analysis of Atorvastatin Calcium Trihydrate API Batch ATC-2026-B09, QC analyst reported Impurity D at 0.28% (w/w), significantly exceeding the validated specification limit of <= 0.15% (w/w). Laboratory phase 1 investigation confirmed no analytical testing error, dilution fault, or instrument drift. Subsequent manufacturing investigation of Agitated Nutsche Filter-Dryer ANFD-02 revealed that during the vacuum drying sequence at 14:15 on 2026-10-22, jacket heating control valve V-104 remained in full-open position despite the batch temperature reaching the recipe target of 40°C. Temperature soared to a peak of 58°C and remained elevated between 52°C and 58°C for approximately 45 minutes before manual intervention by lead operator. Atorvastatin is susceptible to thermal degradation and ester rearrangement at temperatures above 50°C in the presence of trace moisture.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. Batch ATC-2026-B09 was immediately placed on QA Quarantine status in SAP ERP (System Status: Q-LOCKED).
2. Physical yellow quarantine tags were applied to all 12 fiberboard drums located in Warehouse 3 (Bay D-14).
3. SCADA system lock was applied to dryer ANFD-02, preventing further charging or processing until mechanical clearance.
4. Retain samples sent to Analytical R&D; for mass spectrometry structure verification of degradation peaks.

Prepared by:	R. Patel (QC Senior Analyst)	Date: 2026-10-23
Verified by:	M. Vance (Production Manager)	Date: 2026-10-23
QA Approval:	Dr. S. Thornton (Head of Quality Assurance)	**Status: PENDING ROOT CAUSE ANALYSIS**